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Audit Session

ID: 445A53

- OVERVIEW
- METRICS
- FEATURE IMPORTANCE
- GEMINI REPORT
- REMEDIATION

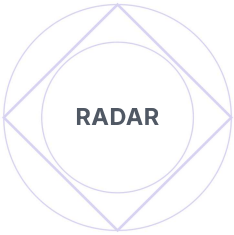
RISK ASSESSMENT

LOW SEVERITY

0 of 5 metrics failed.

Audit Summary

0 of 5 core fairness metrics failed. home_ownership and verification_status are significant predictors of outcome.



FAIRNESS METRICS APR 26, 2026

Demographic Parity Violation

PASS

Measures difference in selection rates across demographic groups.

0.000 vs < 0.1 target

MORTGAGE RENT OWN

Equalized Odds Violation

PASS

Measures difference in true positive and false po

0.0

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Predictive Parity Violation

PASS

Measures difference in precision across groups.

0.000 vs < 0.1 target

MORTGAGE RENT OWN

Disparate Impact

PASS

Ratio of selection rates (min/max). Below 0.8 typically signals disparate impact.

1.000 vs > 0.8 target

MORTGAGE RENT OWN

Individual Fairness Violation

PASS

Proxy based on k-NN consistency. Checks if similar inputs receive similar predictions.

0.000 vs < 0.1 target

MORTGAGE RENT OWN

Outcome Rate by Protected Group

Outcome Rate by Group

Percentage of positive outcomes per protected group

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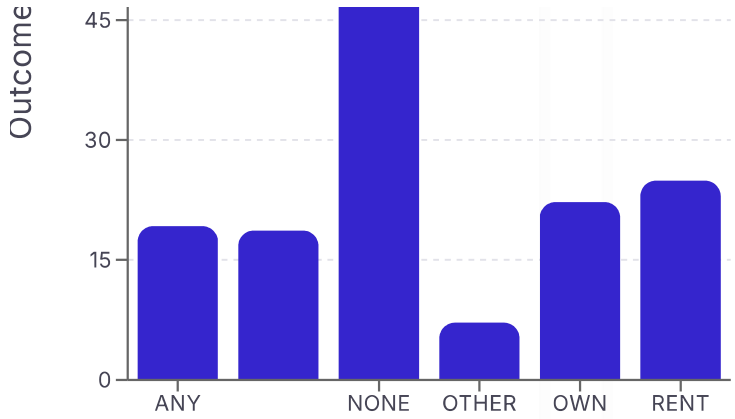

Positive outcome rates per group in the home_ownership protected attribute

What's Driving the Model?

SUB_GRADE	+0.31
TERM	+0.20
DTI	+0.17
INT_RATE	+0.16
ANNUAL_INC	+0.12
INSTALLMENT	+0.07
REVOL_UTIL	+0.07
PURPOSE	+0.05

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REPORT

Audit Findings

SUMMARY

Our review found that the loan default prediction model appears to be very fair. This means it does not seem to unfairly penalize or favor applicants based on their housing situation (e.g., renting vs. owning) or the verification status of their income. The model makes its decisions using appropriate financial information like credit grade, loan length, and debt-to-income ratio, which are directly related to a person's ability to repay a loan. Because the results are exceptionally good—perfect, in fact—we recommend double-checking our testing process to be absolutely certain and setting up ongoing monitoring to ensure the model continues to operate fairly in the future.

WHO IS AFFECTED

No specific groups identified

ROOT CAUSE

The model's fairness appears to stem from its reliance on strong, financially relevant predictors rather than the protected attributes. The top SHAP features ('sub_grade', 'term', 'dti', 'int_rate') are direct indicators of credit risk. The protected attributes 'home_ownership' and 'verification_status' are not among the top drivers of predictions, suggesting the model has learned to predict

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SUGGESTED:

How can I mitigate race proxy bias?

Which group is most affected by this model?

What are the top remediation steps I should take?

Ask Gemini about this au

STRATEGIC REMEDIATION PROTOCOL

01 Validate the audit setup: Given the perfect scores (0.00 violations, 1.00 DI), re-verify the audit tool's configuration and metric calculations to ensure their accuracy and rule out methodological errors.

02 Implement continuous monitoring: Establish a system to track these fairness metrics over time to detect any potential bias that may arise from data drift or future model updates.

03 Investigate proxy variables: Analyze the correlation

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attributes to ensure they are not acting as strong proxies.

04 Expand metric suite: Consider incorporating additional fairness metrics, such as counterfactual fairness or analyzing error rates for specific subgroups, to provide a more holistic view of model performance.

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